The mitochondria consist of a smooth outer membrane, which has a large number of special proteins known as the porins, separated by a space from an inner membrane. The inner membrane is thrown into folds or invagination called cristae which extend into matrix, the mitochondrial lumen. Both the membranes are separated by a clear inter membrane space. The cristae are irregularly shaped like villous and finger like projections. The membranes are made up of phospholipids and proteins.

Functions of Mitochondria

- The mitochondria can help the living cell to convert energy supplied by the environment into ATP, the common molecule, required for chemical reactions. ATP can be generated in two pathways: in the cytosol, and in mitochondrion. First pathway exists in the cytosol of an eukaryotic cell (or within a bacterial cell) where glycolysis degrades glucose to lactate and releases two molecules of ATP.
- Second pathway is the main source of energy production as ATP (called oxidative phosphorylation and involves the electron transport chain). Pyruvate generated from glycolysis enters the matrix (lumen) of the mitochondrion, where it is degraded and combined with coenzyme A to form acetyl CoA. The acetyl part of the acetyl CoA is then degraded to carbon dioxide by the citric acid cycle, releasing hydrogen atoms. The hydrogen atoms are used to reduce the carrier NAD^+ to NADH, and then oxidation of NADH releases a proton and an electron.
- Mitochondria help the cells to maintain proper concentration of calcium ions within the compartments of the cell.
- Mitochondria also help in erythropoiesis and biosynthesis of hormones like testosterone and estrogen.
- The mitochondria of liver cells have enzymes that detoxify ammonia.
- The mitochondria also play an important role in the process of apoptosis or programmed cell death. Abnormal death of cells due to the dysfunction of mitochondria can affect the function of an organ.
- The mitochondria are involved in other cellular activities like signalling, cellular differentiation and cell senescence. They also regulate the control of cell cycle and cell growth.
- Unlike the outer membrane, the inner membrane is strictly permeable, it is permeable only to oxygen, ATP and it also helps in regulating transfer of metabolites across the membrane.
- The matrix of the mitochondria is a complex mixture of proteins and enzymes. These enzymes are important for the synthesis of ATP molecules, mitochondrial ribosomes, tRNAs and mitochondrial DNA.
- Mitochondria also affect human health. Mitochondrial disorders and cardiac dysfunction also play an important role in the aging process.

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